Predicting species richness and diversity

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Questions so far?

Background

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Without a generative model:

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- No ability to account for sampling: effort, detection, protocol
- No ability to account for data properties: overdispersion, zero-inflation
- No prediction for new sites, times, or conditions

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A community model addresses all of these: diversity follows from the generative model and inherits its inferential properties.

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But it is a **derived quantity** from multivariate community data. Analyzing it directly means ditching important aspects of the data.

The typical approach

Species richness is most commonly analyzed with log-linear regression (Poisson or negative-binomial):

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This is simple and interpretable. But it rests on four assumptions that are routinely violated in practice:

1. Species are **independent** of each other
2. All species respond **identically** to the environment
3. The species pool is **infinite** (unbounded distribution)
4. The chosen distribution is **appropriate**

Assumption 1: species independence

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Real communities co-occur non-randomly beyond what the environment explains: facilitation, competition, and shared microhabitat leave residual correlation.

A Poisson or NB regression assumes this away entirely: it has no way to represent residual structure among species, whatever its strength or direction.

Assumption 2: homogeneity of niches

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The consequence: estimated coefficients from a Poisson richness regression are a **weighted average** over species, with weights depending on the distribution of species' niches, usually not the average ecologists want.

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Species richness is inherently bounded by the number of species in the observed pool.

The correct distribution: Poisson-Binomial

Under species independence and Bernoulli responses, species richness follows a **Poisson-Binomial distribution**:

$$SR_i \sim \mathcal{PB}(\pi_{i1}, \pi_{i2}, \dots, \pi_{im}) \tag{3}$$

with

$$\mathbb{E}(SR_i) = \sum_{j=1}^{m} \pi_{ij}, \qquad \mathsf{var}(SR_i) = \sum_{j=1}^{m} \pi_{ij}(1 - \pi_{ij}) \tag{4}$$

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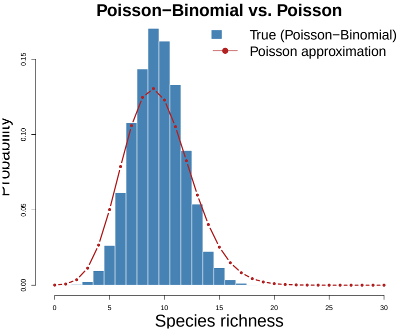
- Mean richness = **sum of species' occurrence probabilities**
- Variance is bounded (cannot exceed $m/4$)
- This is the distribution that actually generates species richness
- The Binomial is a special case (all π_{ij} equal)

Why univariate analysis fails



The Poisson distribution:

- Puts probability mass on impossible values (SR $>$ m)
- Has the wrong shape (underdispersed here)
- Assumes the wrong mean-variance relationship

The mismatch worsens when species niches are heterogeneous.

Species richness as an emergent property

We do not model species richness directly; instead, **species richness emerges** from each species' response to the environment:

$$\mathbb{E}(SR_i) = \sum_{j=1}^{m} \mathbb{E}\{I(y_{ij} > 0)\} = \sum_{j=1}^{m} p(y_{ij} > 0) \qquad (5)$$

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- No separate richness model needed
- Uncertainty propagates naturally from species-level predictions
- Non-linear richness gradients emerge even from linear species responses
- Inherits whatever structure the species model has: traits, phylogeny, spatial or hierarchical effects

Works for any data type

For any distribution with zero support, the probability of presence can be derived:

Data type	Distribution	$p(y_{ij} > 0)$
Binary	Bernoulli	π_{ij}
Count	Poisson	$1 - e^{-\mu_{ij}}$
Count	Negative-binomial	$1 - (1 + \mu_{ij}/k)^{-k}$
Biomass	Tweedie	$1 - \text{Tw}(0; \mu_{ij}, \phi_j, p)$
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The only requirement: the distribution must assign probability mass at zero. Gamma and beta distributions do not.

Implementation in gllvm

The `predictSR()` function computes:

- Expected species richness $\hat{S}_i = \sum_j \hat{p}_{ij}$
- Full Poisson-Binomial distribution at each site
- Prediction intervals; optionally for new predictor values
- Per-species AUC via `goodnessOfFit()`: well-calibrated $\hat{p}_{ij}$ give well-calibrated $\hat{S}_i$

```
sr_pred <- predictSR(model, newX = newdata,
                     newLV = matrix(0, ncol = 2, nrow = nrow(
sr_pred$expected$fit  # expected species richness
residuals(sr_pred, model)   # Dunn-Smyth residuals
goodnessOfFit(Y, object = model, measure = "AUC") # AUC per s
```

Example: alpine plants in Switzerland

- Data by D'Amen et al. (2018)
- Occurrence of 175 species at 840 plots
- Binary data (presence-absence)
- Sampled on an elevation gradient in Switzerland

```
dim(Y)
```

```
## [1] 840 175
```

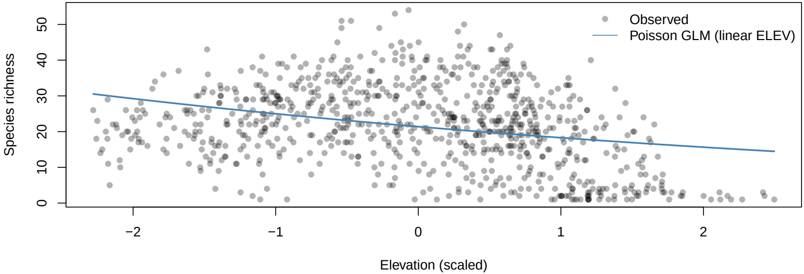
Example: fit a JSDM

```
## gllvm
##    12
## attr(,"autopar")
## [1] TRUE
```

```
model <- gllvm(y = Y[,c(59,(1:ncol(Y))[-59])], X = X_scaled,
               formula = ~ELEV,
               num.lv = 2,
               family = "binomial",
               sd.errors = FALSE)
```

We include elevation as a fixed covariate and two latent variables to capture residual species associations.

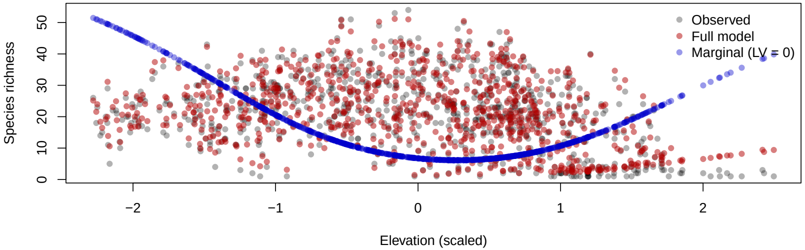
Example: a Poisson GLM for species richness



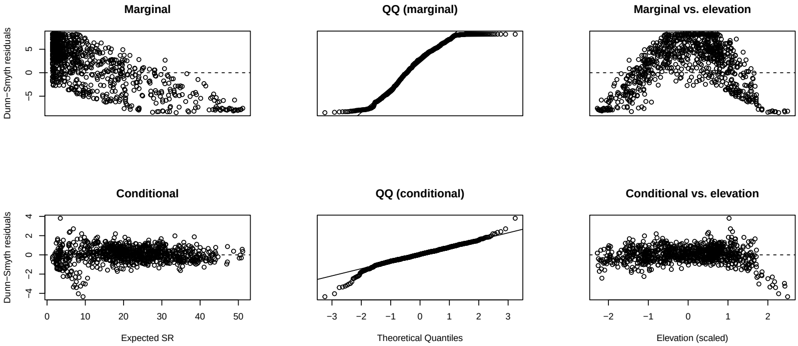
A Poisson GLM with linear elevation predicts a monotone richness curve. Does the multispecies model agree?

Example: LV-elevation confounding

```
obs_sr  <- rowSums(Y)
sr_full <- predictSR(model, se.fit = FALSE)
sr_marg <- predictSR(model, se.fit = FALSE, newLV = matrix(0, ncol = 2, nr
```



Example: SR residuals, linear elevation term



Marginal residuals (LV = 0) reveal the arch the linear fixed effect misses; conditional residuals are flat because the latent variables absorb the curvature.

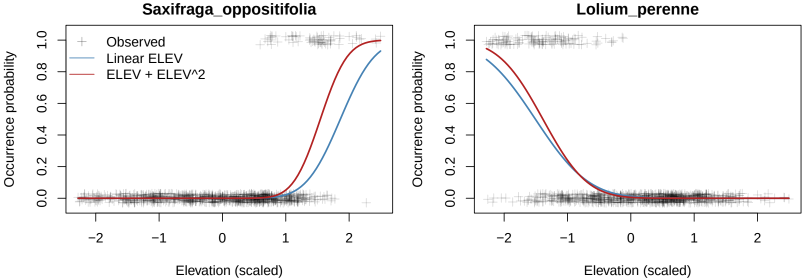
Example: adding a quadratic term

```
model_lin <- model
model <- gllvm(y = Y[,c(59,(1:ncol(Y))[-59])], X = X_scaled,
               formula = ~ELEV + I(ELEV^2),
               num.lv = 2,
               family = "binomial",
               sd.errors = TRUE)
```

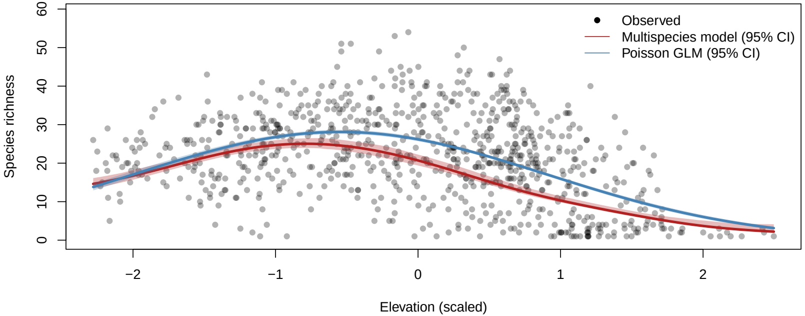
Species richness peaks at intermediate elevation. Without ELEV^2 in the fixed effects, the latent variables absorb the curvature and new-site predictions fail. Adding the quadratic keeps the nonlinearity in the fixed effects where it belongs.

Example: why plot(model) does not reveal this

- `plot(model)` uses **conditional** fitted values: latent variables are estimated per site and adapt to each model
- A missing quadratic term is absorbed by the latent variables; conditional residuals look clean for both models
- Marginal predictions (LV $= 0$) reveal the discrepancy

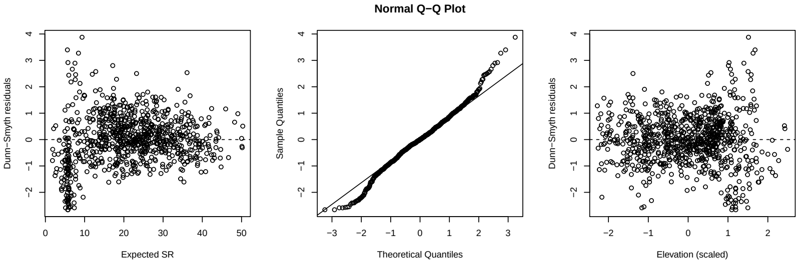


Example: comparing predictions



Question: Both models include a quadratic elevation term. Why do their predictions differ?

Example: residual diagnostics



Poisson-Binomial quantile residuals: no systematic elevation trend remains after adding the quadratic term.

From richness to diversity

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But diversity has many facets: how dominant is the community at a site? How similar are nearby sites? How variable is composition across the region?

These all follow from the **variance structure of** η_{ij}; like R^2 in mixed models (Nakagawa & Schielzeth 2013), we work on the link scale rather than the response scale.

Richness and composition, together

Species richness came from a fitted community model: species-level predictions and any ordination are still there, untouched.

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Richness and composition can therefore be analyzed simultaneously, from one model. They can move independently:

- Richness **unchanged**, composition **changed**: species replace one another (pure turnover)
- Richness **changed**, composition **stable**: species gained or lost without reshuffling the rest

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A model reporting only SR_i cannot tell these apart.

Alpha, beta, and gamma diversity

- **Alpha**: local diversity at a single site; how uneven is the community here?
- **Gamma**: regional diversity; how much variation is there across the whole community?
- **Beta**: turnover between sites; how different are two communities?

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- **Beta**: turnover between sites; how different are two communities?

In a model, these follow from the **variance-covariance structure** of η_{ij}: not new quantities, but different summaries of the same fitted object that also gives you species richness.

Setting up the model

`gllvm` does not (yet) have a dedicated function for alpha, beta, and gamma diversity: what follows is computed directly from a fitted model's variance components. Still worth working through, since the ingredients are already sitting in every model you fit.

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For $\eta_{ij} = \beta_{0j} + \mathbf{x}_i^\top{}_j$ with $\beta_{0j} \sim \mathcal{N}(0, \sigma^2_{\beta_0})$ and ${}_j \sim \mathcal{N}(\mathbf{0}, \Sigma_\beta)$:

- $\text{var}_j(\eta_{ij})$: how much species disagree at site $i \Rightarrow$ **alpha**
- $\text{cov}_j(\eta_{ij}, \eta_{kj})$: how similarly species respond at sites i and k $\Rightarrow$ **beta**
- $\mathbb{E}_i\{\text{var}_j(\eta_{ij})\}$: average disagreement over sites $\Rightarrow$ **gamma**

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The following slides present calculations for a specific model. Results are model-dependent.

Dominance

Dominance is the within-site variance across species:

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- Large D_i: strong environmental filtering; a few well-adapted species dominate
- Small D_i: weak environmental sorting; more even composition

Gamma diversity

Gamma diversity is the expected within-site variation over sites:

$$\gamma = \sigma^2_{\beta_0} + \mathbb{E}_i(D_i) \tag{7}$$

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Regional diversity decomposes into a baseline $\sigma^2_{\beta_0}$ (species differ even at the environmental centroid) and average dominance $\mathbb{E}_i(D_i)$ (the contribution of environmental sorting across sites).

Pairwise similarity (beta diversity)

Pairwise similarity between sites i and k is the covariance of the linear predictor across species:

$$\mathrm{cov}_j(\eta_{ij}, \eta_{kj}) = \sigma^2_{\beta_0} + \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k \tag{8}$$

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- $\sigma^2_{\beta_0}$: baseline similarity shared by all site pairs
- $\mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k$: environmental similarity; large when sites share the same environment

Turnover

Turnover between sites i and k follows from the covariance structure:

$$\begin{aligned}\mathsf{var}_j(\eta_{ij} - \eta_{kj}) &= \mathsf{var}_j(\eta_{ij}) + \mathsf{var}_j(\eta_{kj}) - 2\,\mathsf{cov}_j(\eta_{ij}, \eta_{kj}) \\ &= \sigma^2_{\beta_0} + D_i + \sigma^2_{\beta_0} + D_k - 2\left(\sigma^2_{\beta_0} + \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k\right) \\ &= (\mathbf{x}_i - \mathbf{x}_k)^\top \Sigma_\beta (\mathbf{x}_i - \mathbf{x}_k) \qquad (9)\end{aligned}$$

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The baseline $\sigma^2_{\beta_0}$ cancels. Turnover has the same quadratic form as D_i, evaluated at the difference $\mathbf{x}_i - \mathbf{x}_k$.

Uniqueness

The average pairwise similarity of site i with all other sites is:

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$$\mathrm{uniqueness}_i = \mathbb{E}_k\left\{\mathrm{var}_j(\eta_{ij} - \eta_{kj})\right\} = \sigma^2_{\beta_0} + D_i + \gamma - 2\left(\sigma^2_{\beta_0} + \mathbf{x}_i^\top \Sigma_\beta \bar{\mathbf{x}}\right) \tag{11}$$

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Sites at environmental extremes have high uniqueness: their communities differ most from the regional average, potential refugia, range edges, endemic communities.

Example: Swiss birds

Swiss breeding bird atlas: 56 species at 1000+ sites, with temperature (bio_1), precipitation (bio_12), and slope.

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Fit a JSDM with random species-specific slopes, no LVs:

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model_birds <- gllvm(Y, X, formula = ~(bio_1 + bio_12 + slp | 1),
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```

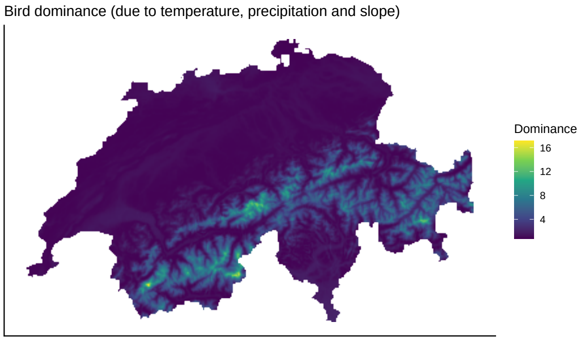
$\hat{\Sigma}_\beta$ is then the estimated covariance of random slopes. All diversity quantities follow from it.

Predicting dominance

```
terra::values(alpha) <- rowSums(x * (x %*% model_birds$params$sigmaB))
```

`model_birds$params$sigmaB` is $\hat{\Sigma}_\beta$: the estimated covariance matrix of random slopes. This evaluates $D_i = \mathbf{x}_i^\top \hat{\Sigma}_\beta \mathbf{x}_i$ at each grid cell: the environmental component of alpha diversity (dominance) due to temperature, precipitation, and slope.

Dominance of Swiss birds



Extending to LVs

Adding $\mathbf{u}_i^\top{}_j$ to the linear predictor introduces residual species associations not captured by environment. Each diversity quantity gains an extra term:

- **Within-site variation**: $\sigma^2_{\beta_0} + D_i + \mathbf{u}_i^\top \frac{1}{m} \Lambda^\top \Lambda\, \mathbf{u}_i$
- **Turnover**: $\Delta\mathbf{x}^\top \Sigma_\beta \Delta\mathbf{x} + \Delta\mathbf{u}^\top \frac{1}{m} \Lambda^\top \Lambda\, \Delta\mathbf{u}$
- **Avg. pairwise similarity**: $\sigma^2_{\beta_0} + \mathbf{x}_i^\top \Sigma_\beta \bar{\mathbf{x}}$, unchanged (LV scores are centered, $\bar{\mathbf{u}} \approx \mathbf{0}$)

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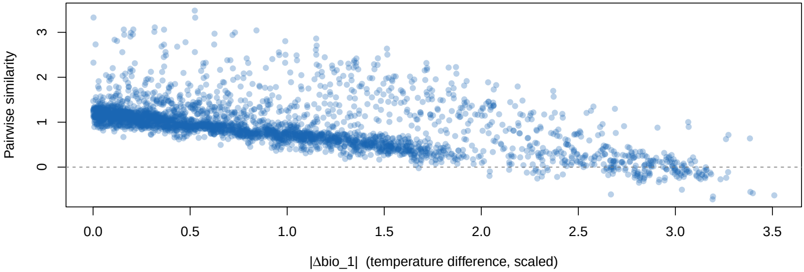
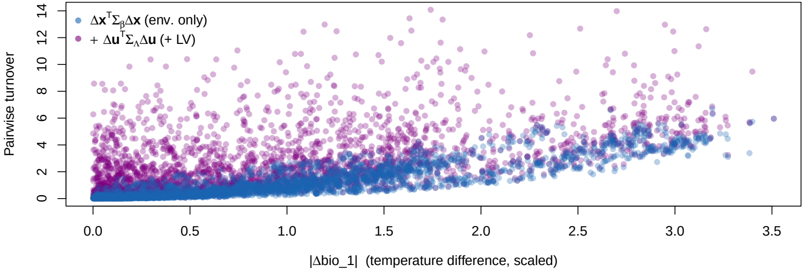
LVs primarily inflate uniqueness (through alpha) and add residual turnover. Gamma and uniqueness extend analogously.

Example: pairwise similarity



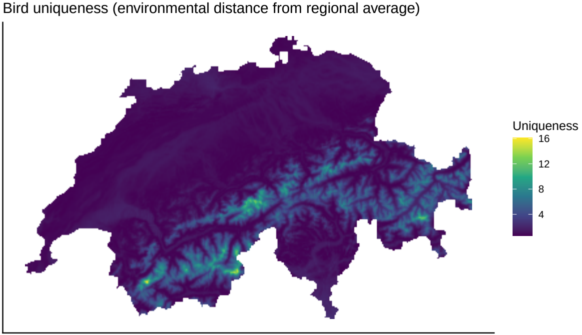
Pairwise similarity decreases with temperature difference. Pairs at opposite ends of the gradient can be negative: their species compositions are anti-correlated, not merely dissimilar.

Example: pairwise turnover



Environmental turnover increases with temperature difference; the LV component adds residual community dissimilarity not explained by the measured predictors. Averaging pairwise turnover over all partners k gives uniqueness: the per-site expected departure from the rest of the landscape.

Example: uniqueness



Sites at environmental extremes are the most unique: their communities differ most from the regional average. These are the sites that drive beta diversity.

Summary

- Model the community; diversity follows
- Univariate species richness analyses make strong and routinely violated assumptions
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Modelling species richness well requires modelling species well first.